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Obtaining ultrafine-grained structure in copper/steel/copper multi-laminated composite by different accumulative roll bonding methods

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ABSTRACT

The relative rollers rotation during roll-bonding affects the effective strain applied to composite layers. In this paper, the differential speed accumulative roll bonding (DSARB) and conventional accumulative roll bonding (CARB) methods were incorporated to fabricate Cu/St/Cu multi-laminated composites. Based on the observation of microstructures, the evolution of microstructure of composites contained necking and rupturing of St layers. In the composite fabricated by DSARB, the St layers showed more irregularities and better distribution in Cu matrix compared to CARB. Based on the observation of grain structures, both methods resulted in ultra-fine-grained structure in layers. However, the DSARB method led to a more pronounced decrement in grain size. The lowest grain size of St and Cu were 2.9 μm and 1.4 μm, respectively, in the DSARB method. Similarly, this method could result in a higher hardness of 401 HV for St and 232 HV for Cu. Also, compared to composite processed by CARB, higher yield and ultimate strength values of 488 MPa and 798 MPa were seen in composite fabricated by DSARB, after seven passes. Furthermore, the fracture surface showed shallower and more sheared dimples in the composite fabricated by DSARB.

1. Introduction

Manufacturing methods play a vital role in creating a new combination of metallic, polymeric, and ceramic materials [1–3]. Therefore, a wide range of methods have been incorporated in recent years [4–9]. In most cases, the methods are selected based on the requirements of industries [10]. One of the interesting metallic materials that offers a diversity of properties is copper [11]. Copper-based alloys and composites have attracted a lot of attention in recent decades [12–17]. Additionally, these materials have been fabricated with the aid of various methods such as powder metallurgy [18], high-pressure die casting [19], additive manufacturing [20], and ultrasonic welding [21].

In recent years, metal matrix composites (MMCs) such as Cu/Al [7], Al/Cu/Ti/Cu/Al [8], and Al/Mg [9] have been synthesized using various methods. The MMCs are mainly produced to obtain a high-strength-to-weight-ratio [22]. The MMCs have many applications such as disk brake, automobile frames, and aircraft engines [23]. Of the processes available for manufacturing MMCs and joining these dissimilar metals, rolling methods such as corrugated rolling [24], ring rolling [25], and roll bonding [8] have been most widely used. The main purpose of each process is to produce sound joints and high-quality composites with the least defects. For example, Xiangyu et al. [26] processed Cu/Al composite by corrugated cold roll bonding. Carpenter et al. [27] processed Cu/Nb by warm rolling. Also, Jiang et al. [28] incorporated vacuum hot rolling in the fabrication of Cu/graphene composite. Similarly, the composites containing steel have been processed by different rolling methods. For example, Fu et al. [29] processed St/Ti composite by cold rolling. Also, Chen et al. [30] fabricated St/Al composite by the hot rolling method.

Many industries such as automotive and aerospace are nowadays seeking economical manufacturing techniques to improve functional properties. Severe plastic deformation methods are extensively used to fabricate multi-laminated composites [31]. One of these methods is the conventional accumulative roll bonding (CARB) process which has repeated passes of cutting, surface-preparing, layer arrangement, and rolling [32]. In recent decades, the effect of rolling pass on different properties of multi-laminated composites has been investigated. Briffod et al. [33] have studied the fatigue and fracture of Cu/Nb composite

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fabricated by CARB. Also, Abdullaeva et al. [34] studied the creep properties of the Al/BN/Cu composite. In addition, Lei et al. [35] reported tensile properties of the Al/Al/Cu composite. The CARB method can be performed at different temperatures, thickness reductions, and speeds. However, the rollers in this method have equal rotational speeds.

In recent years, the differential speed accumulative roll bonding has attracted a lot of attention due to its positive impact on layer deformability, grain refinement, and mechanical. For example, Lee et al. [36] fabricated ultrafine-grained aluminum by differential speed accumulative roll-bonding (DSARB) process. Also, Kim et al. [37] processed pure Ti by high-ratio differential speed rolling. In another work, Polkowski et al. [38] fabricated high-strength nickel thin sheets by differential speed rolling method. It has been reported that higher effective shear strain was obtained in asymmetric accumulative roll bonding compared to conventional symmetric accumulative roll bonding leads to improved deformation rate [39,40]. Also, better grain refinement in layers was caused by grain fragmentation in symmetric accumulative roll bonding [36,39–41]. This is because the applied strain increased as the asymmetric ratio increased.

Up to now, the majority of investigations have been focused on the effect of rolling pass and processing temperature on the microstructure and mechanical properties of composites. To the best of our knowledge, there is limited research work about the role of relative rollers' speed on multi-laminated composites. Therefore, in the current study, the Cu/St/Cu composites were fabricated by differential speed accumulative roll bonding (DSARB) and conventional accumulative roll bonding (CARB) methods. The microstructure, grain size, and mechanical properties of composites were compared.

2. Materials and methods

The three-layer laminate of Cu/medium-carbon steel/Cu which includes two exterior layers of commercially pure copper (99.7 % Cu; 0.02 % Ni; 0.01 % Zn; 0.015 % Fe; 0.002 % Al) and interlayer of medium-carbon steel (0.43 % C; 0.04 % Cr; 0.02 % P; 0.01 % S; 0.2 % Si; 0.75 % Mn; Bal Fe) were utilized in the current research. These metal sheets were selected due to their high ductility which helps them to undergo large deformation. Also, medium-carbon steel was selected as a reinforcing layer because it has higher mechanical properties which help to withstand the transferred load from matrix to reinforcement. The sheets with equal sizes of 100 mm L × 100 mm W × 1 mm T were sliced by using a wire-cutting electric discharge machine.

The DSARB and CARB methods employed seven passes at ambient temperature to avoid the formation of brittle intermetallic phases between layers. Each pass consisted of a 50 % thickness reduction. This thickness reduction was selected to provide sufficient adhesion between layers and to minimize the edge cracks during each pass. In the CARB process, the rolling rotation speed was 2 m/min. In DSARB, the rotation speed of the upper roller was 2 m/min while the rotation speed of the lower roller was 3 m/min, as depicted in Fig. 1. This rolling speed ratio was selected to avoid layer sliding. The microstructures were characterized along the rolling direction by means of scanning electron microscope (SEM) and energy dispersive diffraction (EDS). The grain structure was identified by electron backscattered diffraction (EBSD), and transmission electron microscope (TEM). The indentation tests were conducted by using Vickers microhardness. On the rolling direction (RD)-normal direction (ND) section of each layer, five tests were made. For tensile tests, the samples were cut according to ASTM-E8 standard and then were tested at room temperature. For each composite, three tests were made. After that, the fracture surfaces were observed by SEM.

3. Results and discussion

3.1. Microstructural investigation

The representative SEM micrographs on the RD-ND plane of composites processed by CARB and DSARB methods are shown in Fig. 2. The micrographs seem relatively different in terms of waviness, thickness, necking, and rupture. Usually, the harder layer experiences necking and breakage with an increase in the number of passes.

During the roll-bonding process, the governing mechanisms of the adhesion between layers can be explained based on different possible theories including film theory, energy barrier theory, and penetrative connection theory [42,43]. The contribution of each theory depends on the processing temperature. In this research work, both CARB and DSARB methods were performed at ambient temperature. Therefore, the adhesion mechanism can be described by film theory. According to this theory, the rolling pressure leads to the breakage of surface layers and oxide layers at the interfaces [42]. As a result of this breakage, the microcracking may occur. Through these microcracks, the virgin metals can be extruded and adhered together. The mentioned breakage, extrusion, and adhesion depend on the applied rolling pressure. As the rolling passes increase, more adhesion occurs. It should be noted that energy barrier theory and penetrative connection theory are activated at high-temperature roll bonding. Based on these theories, metallurgical bonding occurs with the aid of intermetallic phase formation [43].

A comparison of the microstructures shows that more irregularities and thickness reduction occurred during DSARB. It is believed that a higher equivalent plastic strain is applied during each cycle of the

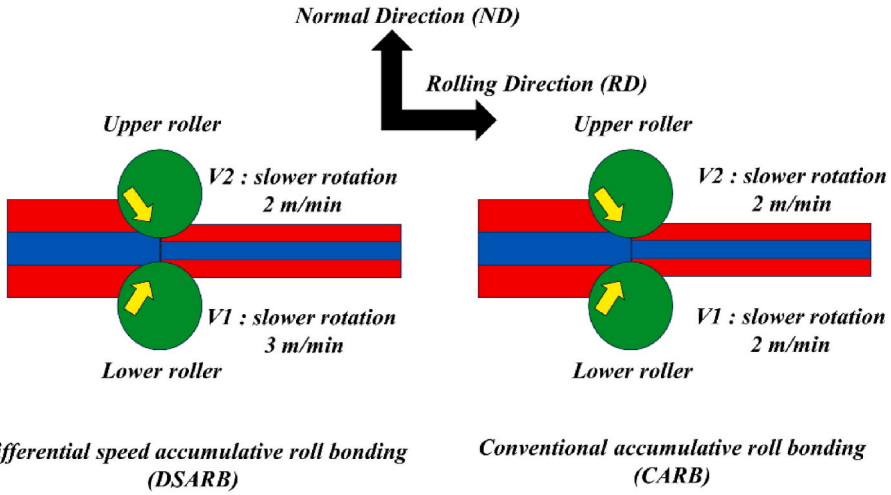


Fig. 1. Illustration of (a) DSARB method with different rotational speeds and (b) CARB method with equal rotational speeds.

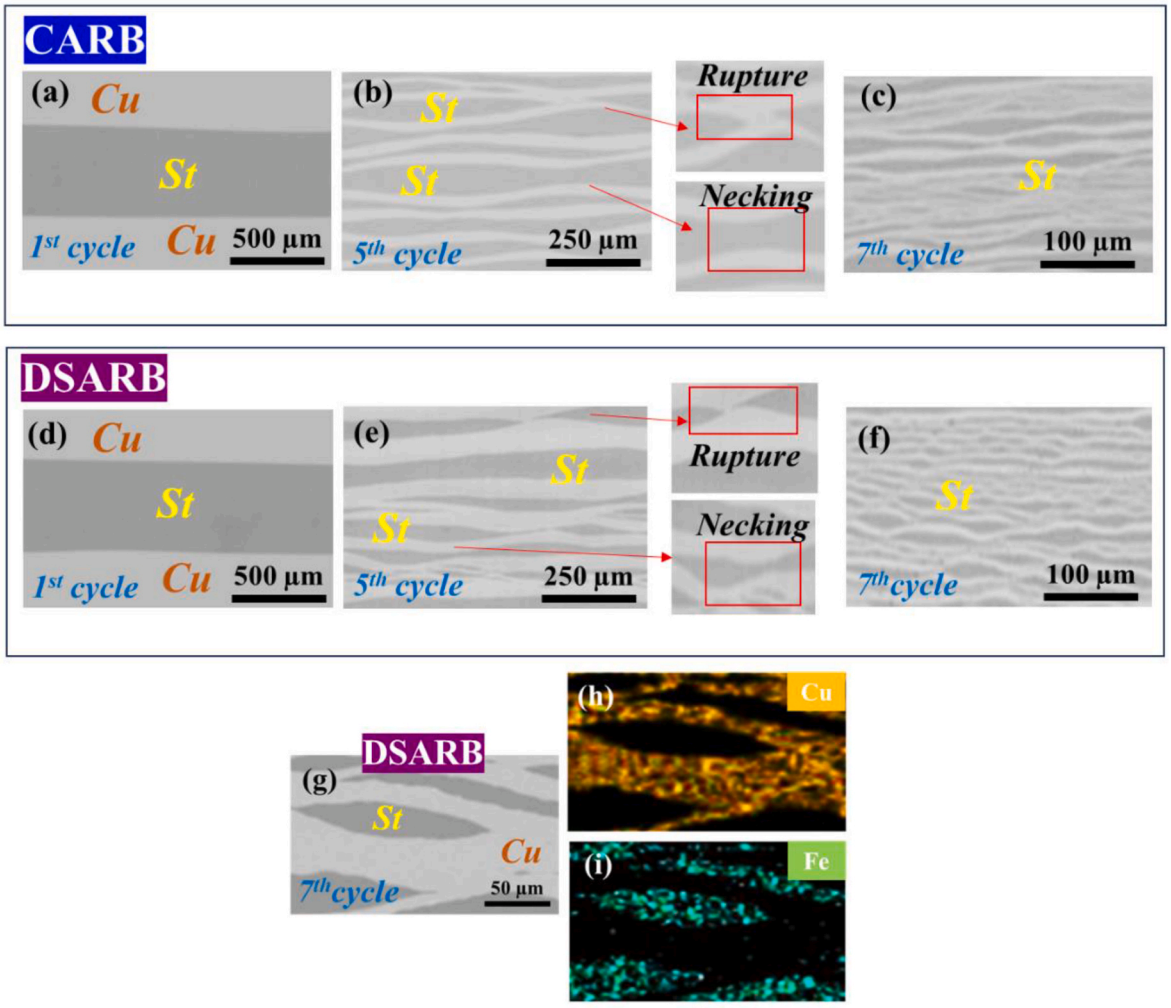


Fig. 2. SEM of Cu/St/Cu composites processed by (a–c) CARB and (d–f) DSARB after (a,d) 1st pass, (b,e) 4th pass, (c,f) 7th pass, and (g–i) EDS analysis.

asymmetric ARB method compared to symmetric ARB [36,39]. However, the wavy pattern exists in both composites. However, the difference in irregularities of microstructures is related to the formation of shear banding. Due to asymmetric roll bonding conditions in DSARB, the shear banding occurs earlier compared to CARB. The shear banding also accelerates the necking and rupturing of layers. In multi-laminated composites, necking and rupturing occur due to different mechanical properties of the matrix and reinforcing layers. It is generally accepted that the load is transferred from the matrix to the reinforcing layer [44]. Here, the St layer as a reinforcement undergoes the transferred load to some extent. With further strain, this layer becomes strain-hardened. When the strain-hardened layer cannot undergo further deformation, it gradually becomes necked. The strain-localization occurs in necked regions. With further strain, the necked regions are separated into small pieces. These pieces are then distributed in the matrix. According to the literature [44,45], necking and rupturing are prevalent in the harder layer. Here, the St layer was harder than the Cu layer and was necked, ruptured, and distributed in the matrix.

The roller-surface friction is the main source of shear gradient. Also, higher effective strain exists on layers in contact with rollers. As shown in Fig. 2g-i, the EDS results confirm that only Cu and Fe elements existed in each separate region. The DSARB method was employed at room temperature so no interdiffusion was observed near interfaces.

3.2. Grain structure

The grain structure evolution of composites for different passes and processes is shown in Fig. 3. The grain refinement was achieved by imposing large strain. The elongated grains in initial passes were gradually replaced by a homogenous fine grain structure by increasing the number of passes. This typical observation has been reported by Dong et al. [46]. Repeated roll bonding is reasonably effective in further reducing the grain size. Here, the DSARB method resulted in finer grains than the CARB method. The minimum grain sizes in St and Cu were 2.9 μm and 1.4 μm, respectively, in the DSARB method while the minimum grain sizes in St and Cu were 3.9 μm and 2.1 μm, respectively, in the CARB method.

Imposing large strain first brings about a high density of dislocation. It is generally accepted that the low-angle grain boundaries and high-angle grain boundaries are created with further plastic deformation [47]. According to TEM images shown in Fig. 4, the grain structure has a lamellar appearance after five passes. The grains are elongated along a rolling direction. By increasing strain, the density of dislocations increases. As a result of dislocation interaction, the subdivision of grains takes place, as shown in Fig. 5. This is associated with the formation of geometrically necessary boundaries (GNBs) and incidental dislocation boundaries (IDBs) [48]. Their interspacing decreases with an increase in a number of passes while the misorientation angles increase. Additionally, GNBs are formed parallel to the rolling direction while the IDBs are perpendicular to the rolling direction. After seven passes, it is clear that

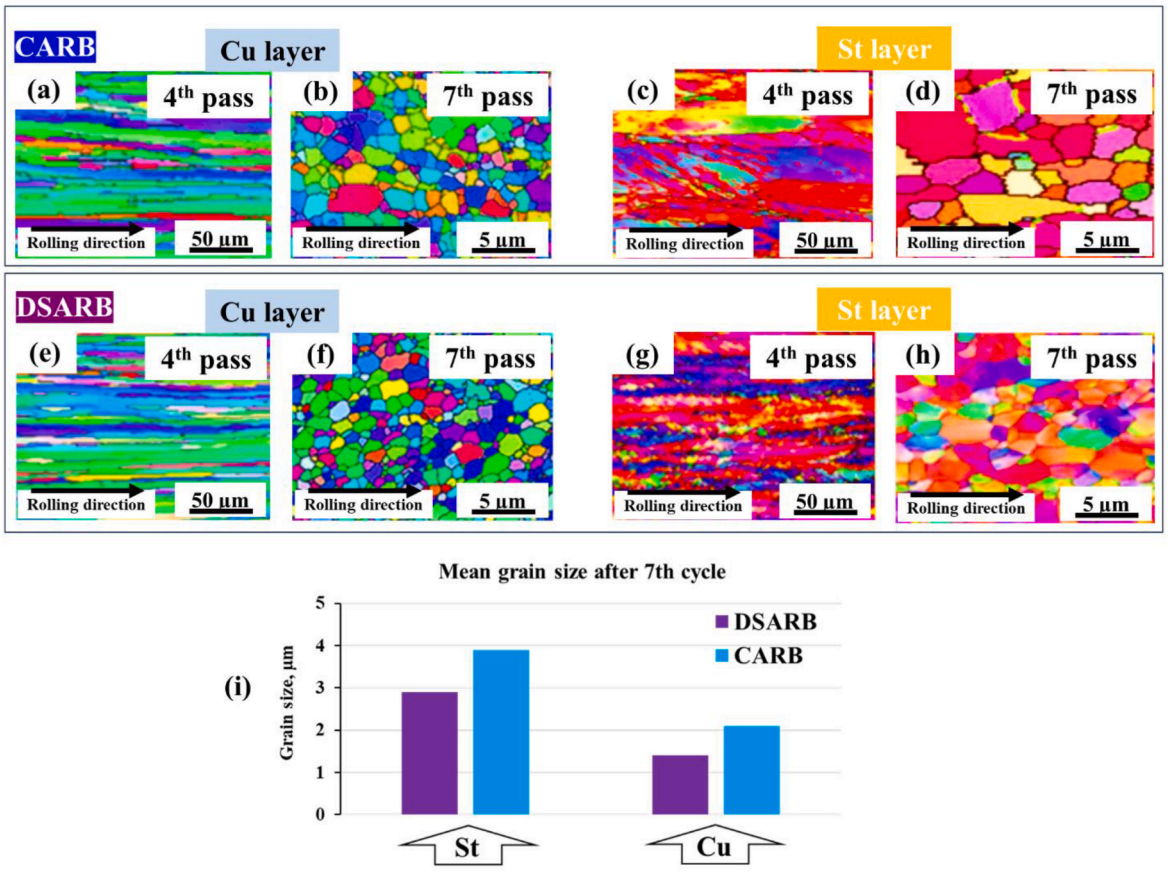


Fig. 3. (a–f) the IPF images of (a,b,e,f) Cu and (c,d,g,h) St layers in composite fabricated by (a–d) CARB method, (e–h) DSARB method after (a, c, e, g) 4th pass, (b, d, f, h) 7th pass, and (i) mean grain size.

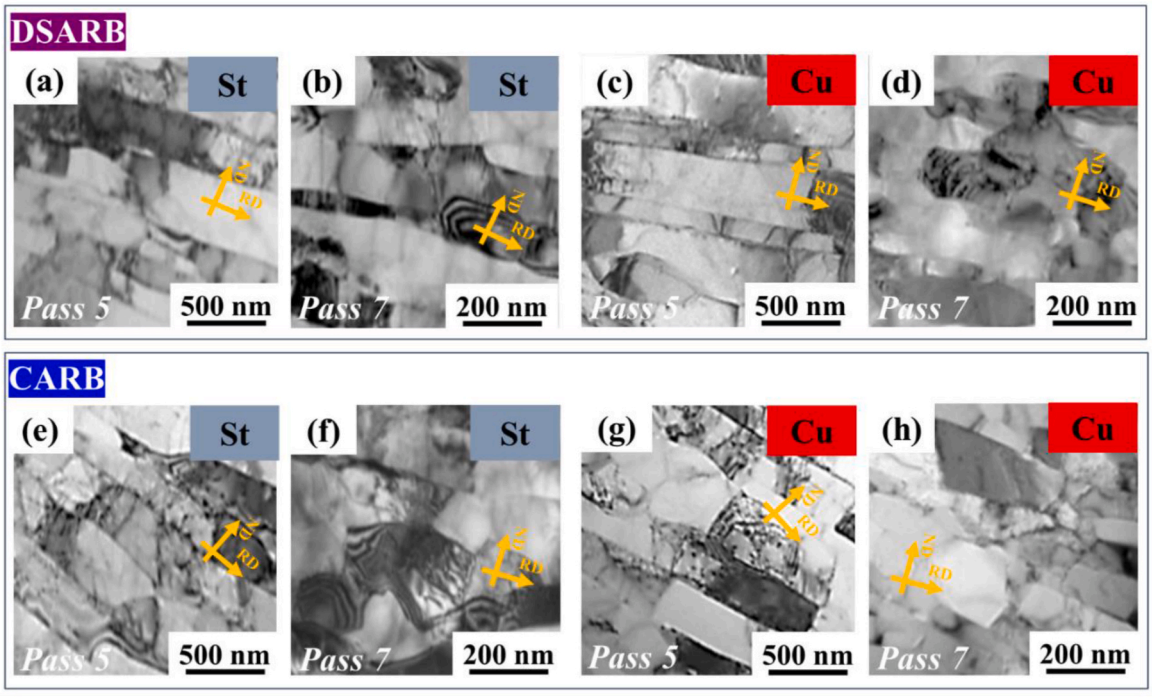


Fig. 4. TEM images of Cu and St layers in composite fabricated by (a–d) CARB method and (e–h) DSARB method.

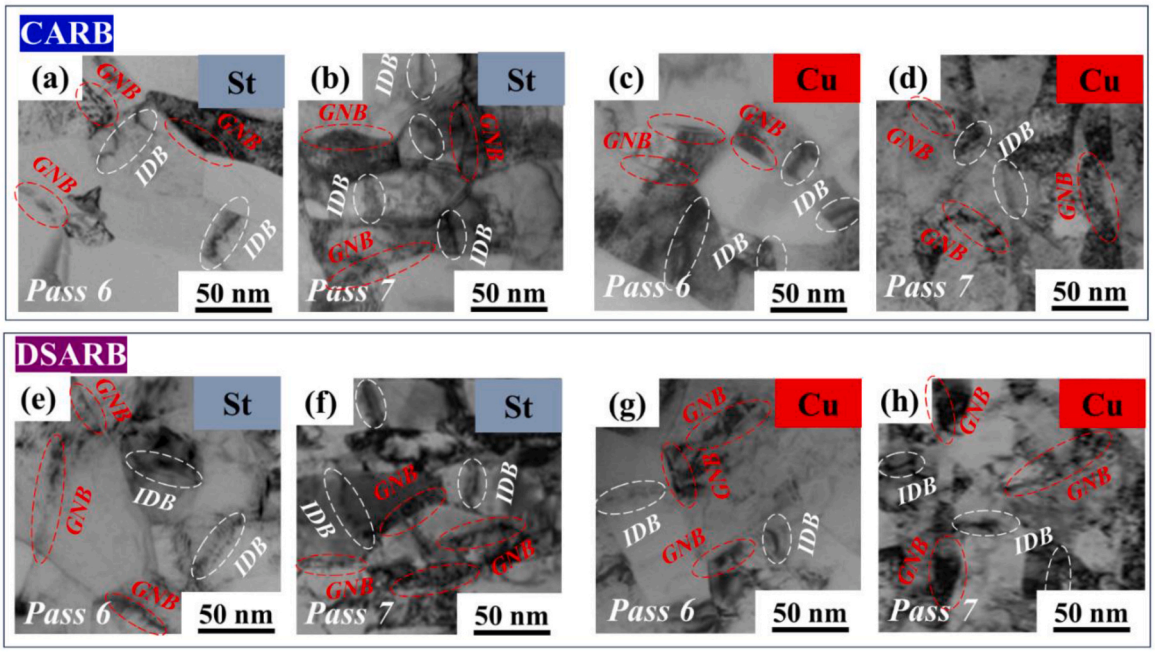


Fig. 5. High-magnification TEM images showing IDBs and GNBs on Cu and St layers in composite fabricated by (a–d) CARB method and (e–h) DSARB method.

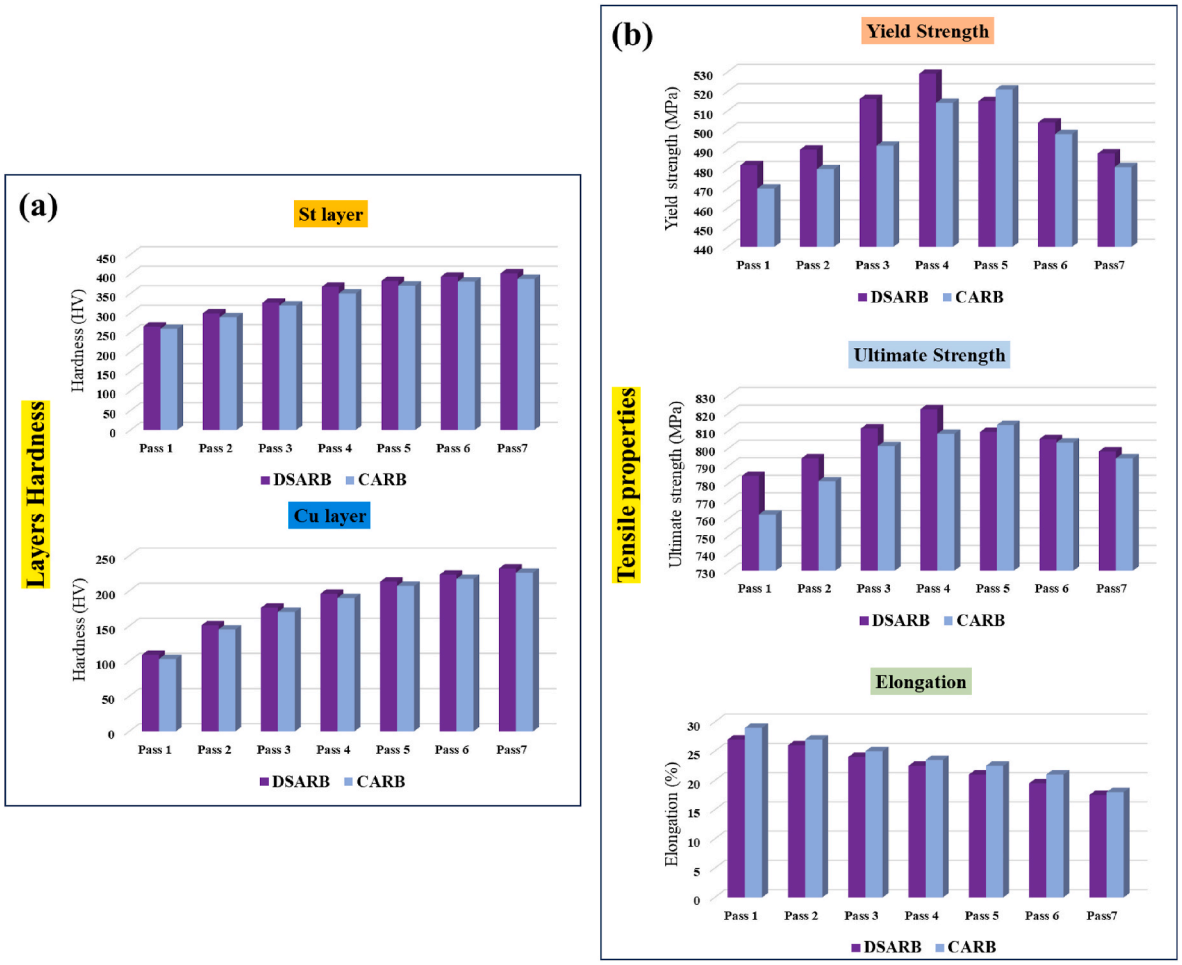


Fig. 6. (a) hardness of St layer and Cu layer, (b) yield strength, ultimate strength, and elongation of composites after different passes of CARB and DSARB methods.

the IDBs formed between GNBs. Therefore, the larger grains were subdivided into smaller ones. Comparing the grain size of layers in composites fabricated by CARB and DSARB indicates larger grains in CARBed composites. The larger effective strain caused by DSARB is reasonably effective in increasing the dislocation density, dislocation interaction, subdivision, and thus further reducing the grain size [36].

3.3. Mechanical properties

The hardness of Cu/St/Cu composites after different passes has been illustrated in Fig. 6a. With an increase in a number of passes, the hardness of the Cu and St layers grew. The indenter brings about a plastically deformed zone on layers. This zone contains a large number of dislocations. These dislocations interact with the dislocation forest in deformed layers. At higher passes, the dislocation tangling is more significant due to the higher density of dislocations. Hence, higher hardness can be expected versus passes. The hardness growth of Cu was more evident than that of St due to the lower stacking fault energy of Cu. The reconfiguration of dislocations in low SFE metals is harder than in high SFE metals. In addition, the DSARB could result in higher hardness values of 232 HV and 401 HV for Cu and St layers, respectively, compared to 226 V and 387 HV for Cu and St layers in the CARBed composite. This is mainly because of applying higher effective strain. This enhancement at higher passes is attributed to its finer grains. In fact, the indenter-induced dislocations interact with more grain boundaries, resulting in higher hardness. During DSARB, the layers are more strain-hardened. This leads to reduced ductility of composites. The increase in hardness values has been reported in composites prepared by CARB in recent years. For example, Jawad et al. [49] reported a 100 HV hardness increment of the St layer during 6 passes in St/Al composite. Also, Tahir and Yapici [50] reported a 40 HV hardness increment of the St layer during 3 passes in the St/Ti composite. As reported by Pita and Lebea [51], the maximum hardness measured in the Cu layer processed by CARB was 116 HV after 2 passes. In another investigation [52], the hardness of Cu reached 160 HV after 8 passes. In comparison to the mentioned investigations, more hardness increment in St and Cu was seen in this study.

Fig. 6b exemplarily illustrates the results from tensile testing for composites processed by different rolling routes. The growth of strengths up to middle passes is seen. This then is followed by the reduction. The reduction of strength is attributed to the discontinuity of harder layers that is responsible for load bearing. The reduction of strength occurs earlier in the DSARBed composite than CARBed composite. This is related to more strain-hardening effect on layers during DSARB process [36]. Because of this, a higher strength of 822 MPa was seen in the DSARBed composite compared to 808 MPa in the CARBed composite, after the 4th pass. The St layers were broken earlier in the DSARBed composite. However, the increase in strength during severe plastic deformation was attributed to strain hardening at low strains [53]. However, in bi-metal composites, the reinforcing layer mainly contributes to strength. When it becomes discontinuous, the composite strength relies on matrix strength. Hence, the strain hardening and grain refinement strengthening mechanisms can be considered for matrix layers at higher passes. In agreement with the literature, the strength of composite increases until when the reinforcing layer is continuous. Tahir and Yapici [50] reported that the strength increased after 3 passes in the Al/Ti/St composite. On the other hand, as reported by Lei et al. [54], in the Al/Al/Cu composite, the strength increased up to the third pass and then decreased until the sixth pass.

The elongation decrease is sharp during initial passes where a large number of dislocations are accumulated. However, composites fabricated by CARB showed relatively better elongation than the other method due to the fact that the layers underwent lower shear strain. Fig. 7a and b exhibit the fracture surfaces of CARB and DSARB samples after tensile tests. All surfaces similarly show a typical ductile fracture. This type of fracture is always accompanied by a large number of dimples of various sizes and depths. The samples are usually fractured as a consequence of microvoids coalescence close to microcrack tips [55]. In addition to dimples, delaminations are present on fracture surfaces. The strain-hardened layers at higher passes show small size of dimples. Comparing the fracture surfaces shows shallower dimples in the DSARBed composite which is probably due to higher applied shear strain during this method. Similar fracture behaviors have been reported in other multi-laminated composites such as AA1100/AA7075 [56], Al/Cu [57], and Cu/Ti [58] in recent years.

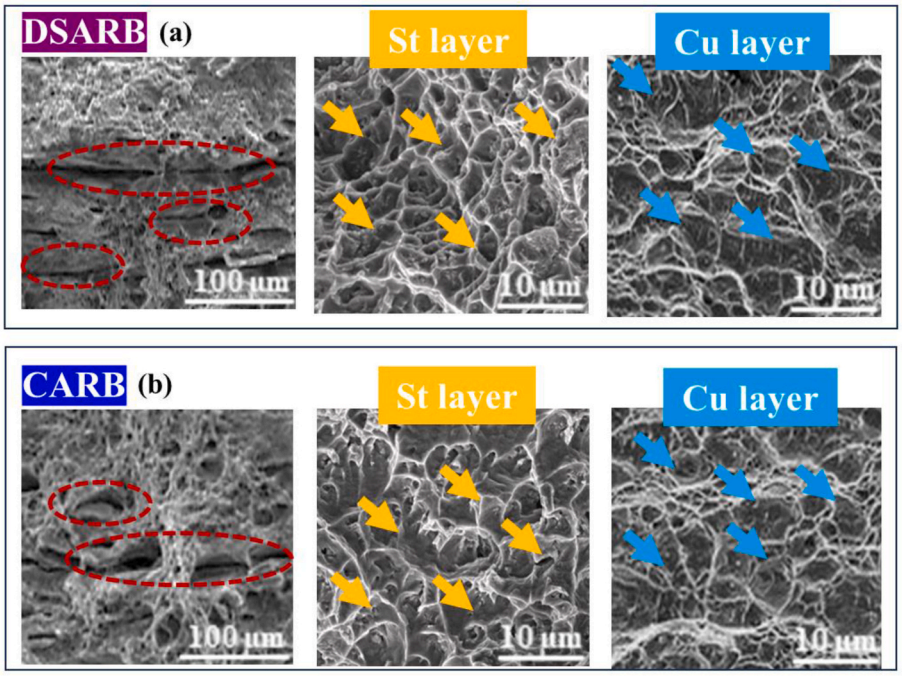


Fig. 7. Fracture surfaces of composite, St, and Cu layers fabricated by (a) DSARBed composite and (b) CARBed composite after seven passes.

4. Conclusion

This research work studied the microstructure, grain refinement, hardness, and tensile properties of Cu/St/Cu multi-laminated composite fabricated by differential speed accumulative roll bonding (DSARB) and conventional accumulative roll bonding (CARB) methods. The summary of main findings is given below.

1. The microstructure of the composite fabricated by DSARB showed more evident necking, fracturing, distribution of the St layer in the Cu matrix than the microstructure of the composite fabricated by CARB.
2. According to EBSD and TEM images, the lowest grain size of St and Cu were 2.9 μm and 1.4 μm, respectively, appeared when the composites were processed by the DSARB method. Also, a more accumulation of dislocations and subdivision was seen in composite layers processed by DSARB.
3. The hardness of the St and Cu layers increased versus passes. The hardness increase in Cu was more pronounced than that of St layer. Also, the DSARB method could result in higher hardness values than the CARB method. The hardness of 232 HV and 401 HV were seen for Cu and St layers after processing by DSARB while the hardness of 226 V and 387 HV for Cu and St layers were seen in CARBed composite, after the 7th pass.
4. The yield and ultimate strengths of composites showed an upward trend up to middle passes which was followed by a downward trend. After the 7th pass, the yield and ultimate strength of 488 and 798 MPa, respectively were obtained by DSARB while the yield and ultimate strength of 481 and 794 MPa, respectively were obtained by CARB. Also, both composites showed ductile fracture although shallower dimples were observed in the DSARBed composite.

CRediT authorship contribution statement

Ailing Ran: Formal analysis, Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing – review & editing. **Ali Afshar:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Shirin Mehmet:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Formal analysis, Data curation. **Yusuf Kemal:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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